

PRACTICAL 1

TOPIC: DDL, DML, DCL, TCL COMMANDS

SECTION 1

DDL:

Data Definition Language is used to define and modify the structure of a database.

Command Purpose

CREATE	Creates a database or table
ALTER	Modifies an existing table
DROP	Deletes a database or table
TRUNCATE	Removes all records from a table
RENAME	Renames a table

```
mysql> CREATE DATABASE bankdb;
Query OK, 1 row affected (0.01 sec)

mysql> USE bankdb;
Database changed
mysql> CREATE TABLE bank_transactions (
    ->     txn_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(50),
    ->     branch_name VARCHAR(50),
    ->     transaction_type VARCHAR(20),
    ->     amount DECIMAL(10,2),
    ->     transaction_date DATE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bankdb |
+-----+
| bank_transactions |
+-----+
1 row in set (0.00 sec)
```

```
mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

```
7 rows in set (0.01 sec)
```

```
mysql> RENAME TABLE bank_transactions
-> TO customer_transactions;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SHOW TABLES;
```

Tables_in_bankdb
customer_transactions

```
1 row in set (0.00 sec)
```

```
mysql> TRUNCATE TABLE customer_transactions;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> SHOW TABLES;
```

Tables_in_bankdb
customer_transactions

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM customer_transactions;
Empty set (0.00 sec)
```

```
mysql> DROP TABLE bank_backup;
ERROR 1051 (42S02): Unknown table 'bankdb.bank_backup'
mysql> CREATE TABLE bank_backup (
    ->     txn_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DROP TABLE bank_backup;
Query OK, 0 rows affected (0.03 sec)
```

SECTION 2

DML— Data Manipulation Language

DML is used to insert, update, delete, and retrieve data from tables.

Command Purpose

INSERT Adds new records
 UPDATE Modifies existing records
 DELETE Removes records
 SELECT Retrieves records

```
mysql> INSERT INTO customer_transactions
    -> (txn_id, customer_name, branch_name, transaction_type, amount, transa
    ction_date)
    -> VALUES
    -> (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
    -> (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
    -> (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
    -> (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
    -> (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
    -> (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
    -> (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
    -> (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
    -> (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15').
```

```
mysql> INSERT INTO customer_transactions
    -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
    -> VALUES
    -> (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM customer_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
111	Venu	Vizag	Deposit	7000.00	2024-01-18	NULL

11 rows in set (0.00 sec)

```
mysql> UPDATE customer_transactions
-> SET amount = 5000
-> WHERE txn_id = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM customer_transactions
-> WHERE txn_id = 105;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> RENAME TABLE customer_transactions TO bank_transactions;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DELETE FROM customer_transactions
-> WHERE txn_id = 111;
ERROR 1146 (42S02): Table 'bankdb.customer_transactions' doesn't exist
mysql> DELETE FROM bank_transactions
-> WHERE txn_id = 111;
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.01 sec)

mysql> SELECT COUNT(*) FROM bank_transactions;
+-----+
| COUNT(*) |
+-----+
| 10 |
+-----+
1 row in set (0.01 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE transaction_type = 'Deposit';
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL

10 rows in set (0.00 sec)

SECTION 3

DCL — Data Control Language

DCL is used to control access and permissions in a database.

Command Purpose

GRANT Gives permissions to users

REVOKE Removes permissions from users

```
mysql> CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.04 sec)

mysql> GRANT SELECT
-> ON bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
+-----+
| Grants for Auditor1@localhost |
+-----+
| GRANT USAGE ON *.* TO 'Auditor1'@'localhost' |
| GRANT SELECT ON `bankdb`.`bank_transactions` TO 'Auditor1'@'localhost' |
+-----+
2 rows in set (0.00 sec)

mysql> CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'manager123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES
-> ON bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
+-----+
| Grants for BranchManager@localhost |
+-----+
| GRANT USAGE ON *.* TO 'BranchManager'@'localhost' |
| GRANT ALL PRIVILEGES ON `bankdb`.`bank_transactions` TO 'BranchManager'@'localhost' |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> REVOKE SELECT
-> ON bank_transactions
-> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
+-----+
| Grants for Auditor1@localhost |
+-----+
| GRANT USAGE ON *.* TO 'Auditor1'@'localhost' |
+-----+
1 row in set (0.00 sec)
```



```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
      -> SET amount = 8000
      -> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.01 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

```

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
      -> SET amount = 99999
      -> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
      -> FROM bank_transactions
      -> WHERE txn_id = 102;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |

```

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
-> SET amount = 7000
-> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
-> SET amount = 9000
-> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
-> FROM bank_transactions
-> WHERE txn_id IN (101,102);
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 7000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> |

```


